

Answers to three problems of Hak, Kozerenko and Oliynyk on the triameter of graphs*

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Abstract

For a connected graph G , the triameter $\text{tr}(G)$ is the largest value of $d(u, v) + d(u, w) + d(v, w)$ over vertices u, v, w . Hak, Kozerenko and Oliynyk asked whether every triametral triple of a median graph contains a peripheral vertex (Problem 1), whether every diametral pair of a median graph extends to a triametral triple (Problem 2), and whether every distance-hereditary graph has at least one of these two properties (Problem 3). For Problem 1 we give an 8-vertex median counterexample and show that it is the unique smallest one. Problem 2 was answered negatively on MathOverflow in December 2025 with an 11-vertex example; we show that the smallest counterexample has 8 vertices, is unique, and is also distance-hereditary. Problem 3 has a positive answer in a stronger form: in a connected distance-hereditary graph, every triametral triple contains a diametral pair or every diametral pair extends to a triametral triple. The proof uses the four-point condition of Bandelt and Mulder. All results of the paper, including the minimality statements and the implication of the theorem of Bandelt and Mulder that the proof uses, are formally verified in Lean 4. The results were found by AI agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

All graphs in this paper are finite and simple. For a connected graph G with path metric $d = d_G$, let $\text{diam}(G)$ be its diameter, put $d(u, v, w) = d(u, v) + d(u, w) + d(v, w)$, and let the *triameter* $\text{tr}(G)$ be the largest value of $d(u, v, w)$ over all $u, v, w \in V(G)$ [2]. A pair $\{u, v\}$ is *diametral* if $d(u, v) = \text{diam}(G)$, a vertex is *peripheral* if it belongs to a diametral pair, and a triple u, v, w is *triametral* if $d(u, v, w) = \text{tr}(G)$. Das [2] asked whether, in a tree, every triametral triple contains a diametral pair and every diametral pair extends to a triametral triple. Hak, Kozerenko and Oliynyk [6] proved both statements for all connected block graphs [6, Theorem 1.2], showed by examples that they fail for other classes, introduced weaker versions, and posed three problems [6, Section 4]. In our notation, a connected graph G has property (Q3) if every triametral triple of G contains a diametral pair;

**Version 2.* This version corrects version 1, which was released on 26 September 2026 in version 1.4.0 of [3]. It adds three hypotheses that version 1 omitted: in Lemma 1 the set S must be nonempty; in the proof of Theorem B, the extension step of Bandelt and Mulder needs a connected graph; and in Section 4, a connected graph needs at least two vertices to have a vertex whose removal leaves it connected. None of these changes Theorems A, B and C. It removes the claims of version 1 that are not formally verified: the counts of connected labelled graphs in Section 4, the last sentence of Remark 5, and Section 6 of version 1, which reported computations on distance-hereditary graphs, apart from its last sentence, which now ends Section 5. Every result of the paper is now formally verified in Lean 4 (Section 6).

(Q3') if every triametril triple of G contains a peripheral vertex;
 (Q4) if every diametral pair $\{x, y\}$ of G extends to a triametril triple, that is, $d(x, y, z) = \text{tr}(G)$ for some vertex z ;
 (Q4') if every peripheral vertex of G belongs to a triametril triple.
 These are Questions 3, 3', 4 and 4' of [6]. We restate the three problems of [6, Section 4] in this notation.

Problem 1. Does every median graph have property (Q3')?

Problem 2. Does every median graph have property (Q4)?

Problem 3. Does every distance-hereditary graph have property (Q3') or property (Q4)?

Both (Q3') and (Q4) hold for trees [6, Theorem 1.2] and for hypercubes, in which every vertex is peripheral and every pair of vertices extends to a triametril triple [6, Corollary 2.5]; and each of them fails for some distance-hereditary graph [6, Figure 3].

Results. Our examples for Problems 1 and 2 are the median graphs G_1 and G_2 on the vertex set $\{0, \dots, 7\}$ in Figure 1; the labels are those of the Lean formalisation.

Theorem A. *Let G_1 be the graph with the ten edges 01, 02, 04, 06, 13, 23, 25, 45, 47, 67. Then G_1 is a median graph with $\text{diam}(G_1) = 4$ and $\text{tr}(G_1) = 8$, its only peripheral vertices are 3 and 7, and $\{1, 5, 6\}$ is a triametril triple. So G_1 does not have property (Q3'), and the answer to Problem 1 is negative. Moreover, every median graph with at most seven vertices has property (Q3'), and G_1 is, up to isomorphism, the only median graph with eight vertices that does not.*

Theorem B (a negative answer to Problem 2 was first given in [13]). *Let G_2 be the graph with the eight edges 02, 03, 06, 12, 25, 34, 35, 67. Then G_2 is a median graph and a distance-hereditary graph with $\text{diam}(G_2) = 4$ and $\text{tr}(G_2) = 12$, and $\{1, 4, 7\}$ is its only triametril triple. So the diametral pair $\{5, 7\}$ does not extend to a triametril triple, and the peripheral vertex 5 belongs to no triametril triple: G_2 has neither property (Q4) nor property (Q4'). Moreover, every median graph with at most seven vertices has property (Q4), and G_2 is, up to isomorphism, the only median graph with eight vertices that does not have (Q4), and also the only one that does not have (Q4').*

Theorem C. *Let G be a connected distance-hereditary graph, and let $\{a, b, c\}$ be a triametril triple of G that contains no diametral pair. Then for every diametral pair $\{x, y\}$ of G ,*

$$\max\{d(a, x, y), d(b, x, y), d(c, x, y)\} = \text{tr}(G).$$

Consequently every connected distance-hereditary graph has property (Q3) or property (Q4), and hence property (Q3') or property (Q4). The answer to Problem 3 is positive.

The last sentences of Theorems A and B rest on an exhaustive enumeration of the median graphs with at most eight vertices (Section 4). Theorem C holds for all finite connected graphs that satisfy the four-point condition of Bandelt and Mulder [1, Theorem 2] (Section 5); Remark 4 shows that it is sharp. All the results, including the enumeration and the two facts from Bandelt and Mulder that we use, are formally verified in Lean 4 (Section 6).

Prior work. The negative answer to Problem 2 is not new. Hak and Kozerenko restated the question in the Lviv Scottish Book [5] (volume 3, page 154, entry dated 10 February 2025), asking whether (Q4), or at least (Q4'), holds for median graphs; the entry also suggests asking whether (Q3') does, which is Problem 1. The problem was relayed to MathOverflow as question 506431 [9] on 28 December 2025, and on 31 December 2025 the user rgvalenciaalbornoz answered it [13], in an accepted answer, with an 11-vertex median graph made of four 4-cycles in which both (Q4) and (Q4') fail. That answer does not concern (Q3'). Our contribution to

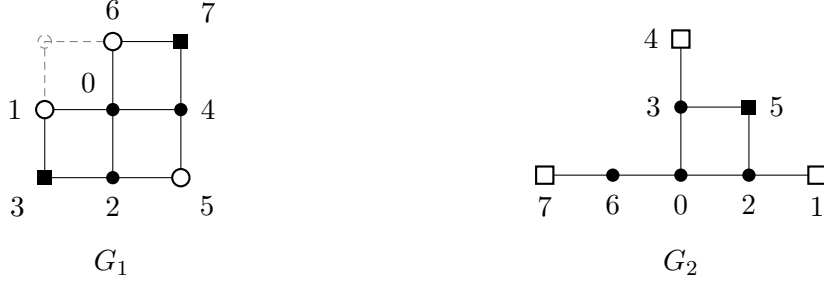


Figure 1: The graphs G_1 and G_2 at the grid points of Section 3. Squares are peripheral vertices. In G_1 the white vertices 1, 5, 6 form a triametal triple without a peripheral vertex; in G_2 the white vertices 1, 4, 7 form the only triametal triple.

Problem 2 is only the smallest counterexample of Theorem B, and the observation that it is also distance-hereditary.

For Problems 1 and 3 we found no earlier answer. We searched on 26 September 2026 between 06:43 and 07:21 UTC: the arXiv (its API, for triameter, triametal, triametric and for the papers of Kozerenko); the works citing [6] listed by OpenAlex and Semantic Scholar, read in full where accessible; Crossref; zbMATH Open; MathDB; MathOverflow and Mathematics Stack Exchange, through the Stack Exchange API; the Lviv Scottish Book; Wikipedia; MathWorld; GitHub (code, commits, issues, pull requests, repositories); the repositories `trureturing` [15], `formal-conjectures` [4] and `SCOPE2026` [14] of AI-assisted work on open problems; A. Hak's bachelor thesis (2022); and web search. The Wikipedia article on the triameter [16], last revised on 19 October 2025, before the MathOverflow answer, lists Problem 2 as open and suggests studying (Q3') and (Q4') for median graphs; the paper [8] (29 August 2026) mentions such questions for median and distance-hereditary graphs without answering them.¹ Some searches failed: the keyword searches of OpenAlex and Semantic Scholar returned HTTP error 429 (rate limit), and we used keyword searches of Crossref instead; one citing paper, in *Symmetry*, was judged from its abstract, as its full text was not accessible to us; and a citation search in zbMATH Open gave no usable result. We did not search Google Scholar, Scopus, Web of Science or ResearchGate. General web search did not find the MathOverflow thread, which we found only through the Stack Exchange API. A final check on 26 September 2026 between 15:49 and 15:52 UTC (the arXiv, GitHub, MathDB and MathOverflow) found nothing new. For version 2 we repeated this final check between 23:38 UTC on 28 September and 00:17 UTC on 29 September 2026 (the arXiv, MathOverflow and Mathematics Stack Exchange, GitHub, MathDB, the works citing [6] listed by Semantic Scholar and OpenAlex, and web search) and found nothing new. This reports the coverage of our searches, not a guarantee of priority.

We quote [6] from its arXiv version 1; its theorem, figure and problem numbers refer to that version.

2 Definitions and readings

Let G be a connected graph with path metric d and $D = \text{diam}(G)$. A vertex u is peripheral if and only if its eccentricity $\max_v d(u, v)$ equals D . As in [6], the vertices in the definition of $\text{tr}(G)$ need not be distinct. For a diametral pair $\{x, y\}$ and any vertex z , the triangle inequality gives $d(x, y, z) \geq 2d(x, y) = 2D$; hence $2D \leq \text{tr}(G) \leq 3D$ [6, Proposition 2.1]. For vertices u, v let $[u, v] = \{x : d(u, x) + d(x, v) = d(u, v)\}$. A connected graph is a *median graph* if $|[u, v] \cap [u, w] \cap [v, w]| = 1$ for all vertices u, v, w [11]; the unique vertex is the *median* of u, v, w .

¹A figure caption in [16] calls the graph G of [6, Figure 3] ($K_{2,3}$ with a pendant vertex) a median graph. It lacks (Q3'), but it is not median, as its vertices a, b, c have two medians; so it does not answer Problem 1.

A connected graph is *distance-hereditary* if every connected induced subgraph H is isometric, that is, $d_H(u, v) = d_G(u, v)$ for all $u, v \in V(H)$ [7].

Readings. A triple with a repeated vertex has $d(u, u, v) = 2d(u, v) \leq 2D \leq \text{tr}(G)$, with equality only if $\text{tr}(G) = 2D$ and $\{u, v\}$ is diametral; so allowing repeated vertices changes nothing in (Q3) and (Q3'). In (Q4), the vertex z may be required to differ from x and y (the *strong reading*) or not (the *weak reading*). If G has at least three vertices the readings agree: if some $z \in \{x, y\}$ works, then $\text{tr}(G) = 2D$, and then every z works; similarly for (Q4'). Our counterexamples refute every reading, and Theorem C holds under the strong one. Also, (Q4) implies (Q4'), since a peripheral vertex x lies in a diametral pair, which extends to a triametral triple containing x .

The sentence of [6] that introduces Problem 2 describes (Q3), while the problem names Question 4. As [6, Figure 4] already shows that (Q3) fails for median graphs, and the restatement [5, 9] concerns (Q4), we read Problem 2 as stated (G_1 also lacks (Q3)). Problem 3 could be read for the class (all distance-hereditary graphs have (Q3'), or all have (Q4)), but the two graphs of [6, Figure 3], shown just before the problem, refute this reading; so we read it for each graph. As both vertices of a diametral pair are peripheral, (Q3) implies (Q3'); so Theorem C gives more than a positive answer, and it also gives (Q3') or (Q4').

3 Two median graphs

We prove Theorems A and B, except for their last sentences, by placing G_1 and G_2 in the integer grid. For $u, v \in \mathbb{Z}^k$ let $\|u - v\| = \sum_i |u_i - v_i|$, and for integers a, b, c let $\text{med}(a, b, c)$ be the middle one of the three. For $S \subseteq \mathbb{Z}^k$ let $\Gamma(S)$ be the graph on S in which u and v are adjacent when $\|u - v\| = 1$.

Lemma 1. *Let $S \subseteq \mathbb{Z}^k$ be finite and nonempty, and suppose that (a) $d_{\Gamma(S)}(u, v) = \|u - v\|$ for all $u, v \in S$, and (b) S is closed under coordinatewise medians: for $u, v, w \in S$, the point $m(u, v, w)$ with coordinates $\text{med}(u_i, v_i, w_i)$ lies in S . Then $\Gamma(S)$ is a median graph, the median of u, v, w is $m(u, v, w)$, and*

$$d(u, v, w) = 2 \sum_{i=1}^k (\max\{u_i, v_i, w_i\} - \min\{u_i, v_i, w_i\}) \quad \text{for all } u, v, w \in S. \quad (1)$$

Proof. By (a), $\Gamma(S)$ is connected. For integers a, b, z we have $|a - z| + |z - b| \geq |a - b|$, with equality if and only if z lies between a and b . Summing over the coordinates and using (a), a point $z \in S$ lies in $[u, v]$ if and only if z_i lies between u_i and v_i for every i . An integer lies between each two of a, b, c if and only if it equals $\text{med}(a, b, c)$: if $a \leq b \leq c$, it must lie between a and b and between b and c , so it equals b , and b does lie between each two. Hence $[u, v] \cap [u, w] \cap [v, w] = \{m(u, v, w)\} \cap S = \{m(u, v, w)\}$ by (b). Finally, if $a \leq b \leq c$ then $|a - b| + |a - c| + |b - c| = 2(c - a)$; applying this in each coordinate and using (a) gives (1). $\square$

The hypothesis that S is nonempty was missing in version 1; it is needed under the convention, which we follow with Mathlib, that a graph without vertices is not connected.

Condition (a) holds if any two points $u, v \in S$ are joined by a *monotone lattice path* in S , that is, a path in $\Gamma(S)$ along which every coordinate changes monotonically. Such a path has length $\|u - v\|$, and no path is shorter, since along each edge the value of $\|x - v\|$ changes by one.

Proof of Theorem A, except the last sentence. Place the vertices of G_1 at the points

$$3 = (0, 0), \quad 2 = (1, 0), \quad 5 = (2, 0), \quad 1 = (0, 1), \quad 0 = (1, 1), \quad 4 = (2, 1), \quad 6 = (1, 2), \quad 7 = (2, 2)$$

(Figure 1). These are the points of $S_1 = \{0, 1, 2\}^2 \setminus \{(0, 2)\}$, and the pairs of them at distance 1 are exactly the ten edges of G_1 ; so $G_1 = \Gamma(S_1)$. Two points of S_1 are joined by a monotone lattice path in $\{0, 1, 2\}^2$. If it passes through the corner $(0, 2)$, it does so between the grid neighbours $(0, 1)$ and $(1, 2)$ of the corner, and replacing $(0, 2)$ by $(1, 1)$ gives a monotone lattice path in S_1 . So (a) holds. For (b), suppose that $m(u, v, w) = (0, 2)$ for some $u, v, w \in S_1$. Then two of u, v, w have first coordinate 0 and two have second coordinate 2; one of them has both, and so equals $(0, 2) \notin S_1$, a contradiction. By Lemma 1, G_1 is a median graph.

The distance $\|u - v\|$ between points of S_1 is at most 4, with equality only for $\{u, v\} = \{(0, 0), (2, 2)\}$, as $(0, 2) \notin S_1$. Hence $\text{diam}(G_1) = 4$, the only diametral pair is $\{3, 7\}$, and the peripheral vertices are 3 and 7. By (1), $d(u, v, w) \leq 2(2 + 2) = 8$, and the vertices $1 = (0, 1)$, $5 = (2, 0)$ and $6 = (1, 2)$ attain this bound, since in each coordinate their values range from 0 to 2. So $\text{tr}(G_1) = 8$, and $\{1, 5, 6\}$ is a triametral triple that contains neither 3 nor 7. Its distances are $d(1, 5) = 3$, $d(1, 6) = 2$ and $d(5, 6) = 3$, so it contains no diametral pair either. (On the other hand, G_1 has (Q4), as $\text{tr}(G_1) = 2 \text{diam}(G_1)$.) $\square$

Proof of Theorem B, except the last sentence. Place the vertices of G_2 at the points

$$7 = (-2, 0), 6 = (-1, 0), 0 = (0, 0), 2 = (1, 0), 1 = (2, 0), 3 = (0, 1), 5 = (1, 1), 4 = (0, 2).$$

The pairs of these points at distance 1 are exactly the eight edges of G_2 , so $G_2 = \Gamma(S_2)$, where S_2 consists of the row $R = \{(t, 0) : -2 \leq t \leq 2\}$ and the points $(0, 1)$, $(1, 1)$ and $(0, 2)$. Two points of R are joined along R ; a point $(t, 0)$ is joined to $(0, 1)$ and to $(0, 2)$ by going along R to $(0, 0)$ and then up, and to $(1, 1)$ by going along R to $(1, 0)$ and then up; and $(1, 1)$, $(0, 1)$, $(0, 2)$ is a monotone path. So (a) holds. For (b), let $u, v, w \in S_2$ and $m = m(u, v, w)$. If at most one of u, v, w lies outside R , then $m_2 = 0$ and $m \in R$. Otherwise two of them lie in $\{(0, 1), (1, 1), (0, 2)\}$, so that $m_1 \in \{0, 1\}$ and $m_2 \in \{1, 2\}$. If $m_2 = 1$, then $m \in S_2$; if $m_2 = 2$, then two of u, v, w equal $(0, 2)$, and $m = (0, 2)$. By Lemma 1, G_2 is a median graph.

The first coordinates of the points of S_2 range over $[-2, 2]$ and the second over $[0, 2]$. So by (1), $d(u, v, w) \leq 2(4 + 2) = 12$, with equality if and only if the triple contains the only point with first coordinate -2 (vertex 7), the only point with first coordinate 2 (vertex 1) and the only point with second coordinate 2 (vertex 4). Hence $\text{tr}(G_2) = 12$, and every triametral triple consists of 1, 4, 7, which are pairwise at distance 4. One checks from the coordinates that no two vertices are at distance more than 4; so $\text{diam}(G_2) = 4$, and $d(5, 7) = \|(1, 1) - (-2, 0)\| = 4$ shows that $\{5, 7\}$ is a diametral pair and 5 is peripheral. As $5 \notin \{1, 4, 7\}$, no triple containing 5 is triametral. So $d(5, 7, z) < 12$ for every vertex z , and G_2 has neither (Q4) nor (Q4'). It has (Q3), since its only triametral triple consists of diametral pairs.

Finally, G_2 is distance-hereditary. Call t a *twin* of a vertex $v \neq t$ if every vertex other than v and t is adjacent to both of them or to neither. We use the easy direction of [1, Theorem 1]: *if G is a connected graph, v is a pendant vertex of G or has a twin in G , and $G - v$ is distance-hereditary, then G is distance-hereditary.* (Version 1 omitted the hypothesis that G is connected, which is needed: two vertices without an edge are twins; deleting one leaves a graph with one vertex, which is distance-hereditary, while the two together are not connected.) Now G_2 is obtained from the edge 02 by attaching a pendant vertex 5 at 2, then adding a vertex 3 adjacent to the two neighbours 0 and 5 of 2 (a twin of 2 that is not adjacent to it), and then attaching pendant vertices 1 at 2, 4 at 3, 6 at 0 and 7 at 6. Every graph in this sequence is connected, and the edge 02 is distance-hereditary; so G_2 is distance-hereditary. $\square$

So (Q4) fails even for graphs that are both median and distance-hereditary; by Theorem C, this can happen only in graphs with property (Q3), like G_2 . The example of [13] is not distance-hereditary: in its notation, the vertices b, v_2, s, w_1, x induce a path of length 4, while $d(b, x) = 2$.

4 The smallest counterexamples

The last sentences of Theorems A and B follow from an exhaustive enumeration of the median graphs with at most eight vertices. It rests on two facts. A median graph has no triangle, since for a triangle u, v, w the intervals $\{u, v\}$, $\{u, w\}$, $\{v, w\}$ have no common vertex. And every connected graph with at least two vertices has a vertex whose removal leaves it connected, such as a leaf of a spanning tree (version 1 omitted the words “with at least two vertices”).

Let G be a connected triangle-free graph with $n \geq 2$ vertices, and let v be a vertex whose removal leaves G connected. Then $G - v$ is connected and triangle-free, and the neighbours of v form a nonempty independent set, as G is connected and has no triangle. So every connected triangle-free graph with n vertices arises from one with $n - 1$ vertices by adding a vertex adjacent to a nonempty independent set. Starting from the graph with one vertex, this gives all connected triangle-free graphs with at most seven vertices; up to isomorphism, there are 59 of them with seven vertices. In particular, every median graph with eight vertices is isomorphic to a graph obtained from one of these 59 graphs by adding a vertex adjacent to a nonempty independent set, and over the 59 graphs there are 1,857 such pairs of a graph and a nonempty independent set, so 1,857 graphs to test. The median graphs with at most seven vertices are among the connected triangle-free graphs with at most seven vertices.

Testing the median property, and properties (Q3'), (Q4) and (Q4'), on all these graphs gives the last sentences of Theorems A and B. Here, as in the rest of the paper, (Q4) and (Q4') have their weak reading, which is their definition (in the strong reading, the median graphs with one or two vertices fail (Q4) trivially). The same enumeration shows that the numbers of median graphs with $n = 1, \dots, 8$ vertices, up to isomorphism, are 1, 1, 1, 3, 4, 11, 23, 69, the first terms of sequence A292623 of the OEIS [12] (median graphs on n nodes). Of the 1,008,904 labelled median graphs on $\{0, \dots, 7\}$, exactly 20,160 lack (Q3') and exactly 20,160 lack (Q4). These are the labelled copies of G_1 and of G_2 , respectively: each of G_1 and G_2 has exactly two automorphisms, and $8!/2 = 20,160$.

This finite check was carried out by computer. For version 1, two independent programs did it, a C program over all graphs with at most eight vertices and a Python program that used the reduction above for eight vertices; their code and logs are in version 1.4.0 of the repository [3], directory `certificates/hko-triameter/`. For version 2 the check is part of the Lean proof (Section 6), where every step, from the reduction to the final counts, is checked by the Lean kernel.

5 Distance-hereditary graphs: proof of Theorem C

Bandelt and Mulder [1, Theorem 2] proved that, for a connected graph G , the following conditions, among others, are equivalent (we keep their numbering). (i) G is distance-hereditary. (vii) For any four vertices u, v, w, x , at least two of the three distance sums $d(u, v) + d(w, x)$, $d(u, w) + d(v, x)$ and $d(u, x) + d(v, w)$ are equal. (viii) G satisfies (vii), and if the two smaller of these sums are equal, then the largest one exceeds them by at most 2. The characterisation quoted in [6, Section 3.2] is (i) $\Leftrightarrow$ (vii); we need (viii). For integers A, B, C write $\text{FP}(A, B, C)$ if two of them are equal and the third is at most their common value plus 2. This is symmetric in A, B, C , and three sums satisfy it exactly when they satisfy the requirements of (viii): if the two larger sums are equal, the third is at most their value, and if the two smaller ones are equal, (viii) bounds the largest. A connected graph G satisfies the *four-point condition* if $\text{FP}(d(u, v) + d(w, x), d(u, w) + d(v, x), d(u, x) + d(v, w))$ holds for all $u, v, w, x \in V(G)$. By the implication (i) $\Rightarrow$ (viii) of [1, Theorem 2], every connected distance-hereditary graph satisfies the four-point condition. This is the only fact about distance-hereditary graphs that we use in the proof of Theorem C. We do not repeat the proof of [1]. The Lean proof of this implication (Section 6) is independent of it: it proceeds by induction on the number of vertices, deleting

a vertex other than the four given ones, with a separate argument when that vertex is a cut vertex, and it checks the graphs with four vertices directly.

The core of the proof is a statement about integers, in which a, b, c are three labels.

Lemma 2. *Let $D, e_{ab}, e_{ac}, e_{bc}, p_a, p_b, p_c$ and q_a, q_b, q_c be integers, put $e_{vu} = e_{uv}$ and $T = e_{ab} + e_{ac} + e_{bc}$, and suppose that for all distinct $u, v \in \{a, b, c\}$*

$$(E) \quad e_{uv} \leq D - 1, \quad (F) \quad \text{FP}(D + e_{uv}, p_u + q_v, p_v + q_u), \quad (N) \quad D + p_u + q_u \leq T - 1.$$

These conditions are inconsistent.

Proof. Put $s_u = p_u + q_u$ and $t_u = p_u - q_u$. Fix distinct $u, v \in \{a, b, c\}$, and let w be the third label. By (F), at least one of the following holds:

$$\begin{aligned} (A_{uv}) \quad & D + e_{uv} = p_u + q_v; & (A_{vu}) \quad & D + e_{uv} = p_v + q_u; \\ (C_{uv}) \quad & p_u + q_v = p_v + q_u \quad \text{and} \quad D + e_{uv} \leq p_u + q_v + 2. \end{aligned}$$

Claim 1. *If (A_{uv}) holds, then $q_v - q_u \geq 2D + 1 - e_{uw} - e_{vw}$ and $t_u - t_v \geq 6$. Put $k = 2D + 1 - e_{uw} - e_{vw}$; by (E), $k \geq 3$. Substituting $p_u = D + e_{uv} - q_v$ into (N) for u gives $2D + e_{uv} - q_v + q_u \leq e_{uv} + e_{uw} + e_{vw} - 1$, that is, $q_v - q_u \geq k$. Substituting $q_v = D + e_{uv} - p_u$ into (N) for v gives, in the same way, $p_u - p_v \geq k$. Adding, $t_u - t_v = (p_u - p_v) + (q_v - q_u) \geq 2k \geq 6$.*

Claim 2. *If (C_{uv}) holds, then $t_u = t_v$ and $2D + 2e_{uv} \leq s_u + s_v + 4$. The equation $p_u + q_v = p_v + q_u$ says that $t_u = t_v$. Its two sides add up to $s_u + s_v$, so both equal $(s_u + s_v)/2$, and the inequality in (C_{uv}) becomes $2D + 2e_{uv} \leq s_u + s_v + 4$.*

By the two claims, for each pair the sign of $t_u - t_v$ decides which case holds: (C_{uv}) if $t_u = t_v$, (A_{uv}) if $t_u > t_v$, and (A_{vu}) if $t_u < t_v$. (For instance, if $t_u > t_v$, then (C_{uv}) fails by Claim 2, and (A_{vu}) fails by Claim 1 with u and v exchanged.) We distinguish cases by the number of pairs $\{u, v\}$ with $t_u = t_v$.

Three pairs. Adding the inequalities of Claim 2 for the three pairs gives $6D + 2T \leq 2(s_a + s_b + s_c) + 12$. By (N), $s_u \leq T - D - 1$ for each u , so $6D + 2T \leq 6T - 6D + 6$, that is, $4T \geq 12D - 6$. But (E) gives $T \leq 3D - 3$, so $4T \leq 12D - 12$, a contradiction.

Two pairs. This is impossible: if $t_u = t_v$ and $t_v = t_w$, then $t_u = t_w$ as well.

One pair, say $\{u, w\}$, and let v be the third label; so $t_u = t_w \neq t_v$. Exchanging the roles of p and q preserves (E), (F) and (N), as FP is symmetric, and replaces every t by $-t$. So we may assume that $t_u = t_w > t_v$. Then (A_{uv}) and (A_{vw}) hold, that is, $p_u = D + e_{uv} - q_v$ and $p_w = D + e_{vw} - q_v$, and hence $s_u + s_w = 2D + e_{uv} + e_{vw} - (q_v - q_u) - (q_v - q_w)$. By Claim 1, $q_v - q_u \geq 2D + 1 - e_{uw} - e_{vw}$ and $q_v - q_w \geq 2D + 1 - e_{uw} - e_{uv}$. Therefore $s_u + s_w \leq 2T - 2D - 2$, and Claim 2 for the pair $\{u, w\}$ gives $2D + 2e_{uw} \leq s_u + s_w + 4 \leq 2T - 2D + 2$, that is, $e_{uv} + e_{vw} \geq 2D - 1$. This contradicts (E).

No pair. Then t_a, t_b, t_c are distinct; call the labels 1, 2, 3 so that $t_1 > t_2 > t_3$. Then (A_{12}) , (A_{23}) and (A_{13}) hold: $D + e_{12} = p_1 + q_2$, $D + e_{23} = p_2 + q_3$ and $D + e_{13} = p_1 + q_3$. Hence $s_2 = p_2 + q_2 = (D + e_{23} - q_3) + (D + e_{12} - p_1) = D + e_{12} + e_{23} - e_{13}$, and (N) for the label 2 gives $2D + e_{12} + e_{23} - e_{13} \leq e_{12} + e_{13} + e_{23} - 1$, that is, $2e_{13} \geq 2D + 1$. This contradicts (E). $\square$

Proposition 3. *Let G be a finite connected graph that satisfies the four-point condition, let $\{x, y\}$ be a diametral pair of G , and let a, b, c be vertices such that $d(a, b)$, $d(a, c)$ and $d(b, c)$ are less than $\text{diam}(G)$. Then $\max\{d(a, x, y), d(b, x, y), d(c, x, y)\} \geq d(a, b, c)$.*

Proof. Put $D = \text{diam}(G) = d(x, y)$ and, for $u, v \in \{a, b, c\}$, $e_{uv} = d(u, v)$, $p_u = d(x, u)$ and $q_u = d(y, u)$; then $T = d(a, b, c)$. Condition (E) of Lemma 2 is the hypothesis. The three distance sums of the vertices x, y, u, v are $d(x, y) + d(u, v) = D + e_{uv}$, $d(x, u) + d(y, v) = p_u + q_v$ and $d(x, v) + d(y, u) = p_v + q_u$, so the four-point condition gives (F). If the conclusion failed, then $d(u, x, y) = D + p_u + q_u \leq T - 1$ for each $u \in \{a, b, c\}$, which is (N). This contradicts Lemma 2. $\square$

Proof of Theorem C. Let G be a connected distance-hereditary graph, or more generally a finite connected graph that satisfies the four-point condition, and let $\{a, b, c\}$ be a triametal triple that contains no diametral pair. Its three distances are at most $\text{diam}(G)$ and not equal to it. For a diametral pair $\{x, y\}$, Proposition 3 gives $\max_u d(u, x, y) \geq d(a, b, c) = \text{tr}(G)$, where u ranges over a, b, c , and the reverse inequality holds by the definition of $\text{tr}(G)$. This proves the displayed equation. If G lacks (Q3), some triametal triple contains no diametral pair, and then every diametral pair extends to a triametal triple, so G has (Q4). As (Q3) implies (Q3'), this proves Theorem C under the weak reading of (Q4).

For the strong reading, let $\{x, y\}$ be a diametral pair. If $\text{tr}(G) = 2 \text{diam}(G)$, every vertex $z \notin \{x, y\}$ extends $\{x, y\}$ (Section 2), and such a vertex exists, since a triametal triple without a diametral pair has three distinct vertices. If $\text{tr}(G) > 2 \text{diam}(G)$, the vertex $u \in \{a, b, c\}$ with $d(u, x, y) = \text{tr}(G)$ is neither x nor y , as $d(x, x, y) = d(y, x, y) = 2 \text{diam}(G)$. $\square$

Remark 4. (1) Neither property suffices alone: the graph G of [6, Figure 3] is distance-hereditary and lacks (Q3'), and the graph H of [6, Figure 3] and G_2 are distance-hereditary and lack (Q4). (2) The hypothesis on $\{a, b, c\}$ cannot be dropped: in H , of diameter 2 and triameter 6, the only triametal triple $\{a, b, c\}$ has pairwise distances 2, and the diametral pair $\{x, y\}$ does not extend. (3) Some hypothesis on G is needed: the graph of [6, Figure 2], of diameter 5 and triameter 12, has only one triametal triple $\{a, b, c\}$, with pairwise distances 4, and only one diametral pair $\{x, y\}$, which does not extend [6, Section 3.2]; as x, y are its only peripheral vertices, it has none of (Q3), (Q3') and (Q4).

Remark 5. As (Q4) holds trivially when $\text{tr}(G) = 2 \text{diam}(G)$, one might try to show that distance-hereditary graphs with $\text{tr}(G) > 2 \text{diam}(G)$ have (Q3'). This is false. Let H_{11} be the graph on $\{0, \dots, 10\}$ with the edges 01, 02, 03, 05, 14, 16, 24, 27, 34, 38, 49 and $\{5, 10\}$: the graph $K_{2,3}$ with parts $\{0, 4\}$ and $\{1, 2, 3\}$, pendant vertices 6, 7, 8, 9 at 1, 2, 3, 4, and a path 0, 5, 10. It is distance-hereditary: it is obtained from the edge 01 by attaching a pendant vertex 4 at 1, adding two twins 2 and 3 of 1 (each adjacent to 0 and 4), and attaching pendant vertices 5 at 0, 10 at 5, and 6, 7, 8, 9 at 1, 2, 3, 4; every graph in this sequence is connected, so the statement used in the proof of Theorem B applies at each step. Its only diametral pair is $\{9, 10\}$, at distance 5; the vertices 6, 7, 8 are pairwise at distance 4; and one checks that $\text{tr}(H_{11}) = 12 > 10$. So $\{6, 7, 8\}$ is a triametal triple without a peripheral vertex. As Theorem C predicts, (Q4) holds: $\{9, 10\}$ extends by each of 6, 7, 8, for example $d(9, 10, 6) = 5 + 3 + 4 = 12$.

Finally, G_1 and the graph of [6, Figure 2] are not distance-hereditary. In G_1 , the vertices 3, 1, 0, 4, 5 induce a path of length 4, while $d(3, 5) = 2$; the graph of [6, Figure 2] has neither (Q3) nor (Q4), which by Theorem C is impossible for a distance-hereditary graph.

6 Formal verification

All the results are formally verified in Lean 4 (version 4.34.1) with Mathlib [10] (version v4.34.1, commit d13f23b7). Version 1 of this paper was released with a partial formalisation, the modules `HKOTriameter.lean` and `HKOTriameterDH.lean` of version 1.4.0 of [3], which covered the two counterexamples and Theorem C, in its weak reading, for graphs that satisfy the four-point condition. For version 2 every result of the paper was formalised: each of its mathematical claims is a statement of Table 1, a definition, a cited result or a step of a written proof.

Statements before proofs. Here the statements were written after the written proofs: they formalise a paper that had already been released. After version 1 had been released, an agent wrote a Lean file of definitions and statements, the *statement file*, which proves none of its statements. An independent agent reviewed version 1 of this file against the paper and the cited sources and checked its definitions with programs of its own. Version 2

of the file applies the review’s corrections, adds the count 1,857 of Section 4, and leaves out the claims removed from this version of the paper. A second AI system reviewed version 2, and it was frozen on 27 September 2026 at 22:04 UTC by recording its SHA-256 hash 9f1b0c95f35addf03f54c330ea0cd63324154d17e1cc8e707c8bf830da4a0785. The proofs were written after the freeze, in separate files that import the statement file, which was not changed. The definitions of the metric notions, of the properties and problems, of the four-point condition and of the graphs G_1 , G_2 , G and H are copied verbatim from the modules of version 1, so that the earlier Lean results remain linked to them.

What is proved. The statement file states 50 propositions, listed in Table 1; the file of checks (Research/Backfill/Paper4/Check.lean) proves each of them, by a theorem `check_X:X` for every name X of the table. In ASCII notation (`forall`, `->`, `/\`, `\|` and `Not`), the definition of distance-hereditary graphs and three of the statements read as follows; `G.dist` and `G.induce` are Mathlib’s distance and induced subgraph, and `SimpleGraph.Iso G G1` is Mathlib’s type of graph isomorphisms.

```
def IsDistanceHereditary (G : SimpleGraph V) : Prop :=
  G.Connected /\ forall s : Set V, (G.induce s).Connected ->
    forall u v : s, (G.induce s).dist u v = G.dist u v
def BandeltMulder_DH_fourPoint : Prop := forall (V : Type) [Fintype V]
  (G : SimpleGraph V), IsDistanceHereditary G -> FourPointBM G
def TheoremC_ii : Prop := forall (V : Type) [Fintype V] (G : SimpleGraph V),
  IsDistanceHereditary G ->
    (Question3 G \| Question4 G) /\ (Question3' G \| Question4 G) /\
    (Question3 G \| Question4Strong G) /\ (Question3' G \| Question4' G)
def TheoremA_unique : Prop := forall (V : Type) [Fintype V] (G : SimpleGraph V),
  Fintype.card V = 8 -> IsMedian G -> Not (Question3' G) ->
    Nonempty (SimpleGraph.Iso G G1)
```

Table 1: The 50 statements of the frozen statement file (namespace `BackfillPaper4.Challenge`). Each is proved. The names follow the section numbers of version 1: its Sections 6 and 8 are the last paragraph of Section 5 and Section 7 here.

Lean statement	what it says	here
<code>Sec2_peripheral_iff</code>	eccentricity = $\text{diam}(G)$ if and only if the vertex lies in a diametral pair	§2
<code>Sec2_bounds</code>	$2 \text{diam}(G) \leq \text{tr}(G) \leq 3 \text{diam}(G)$, and	§2
<code>Sec2_repeated</code>	$d(x, y, z) \geq 2 \text{diam}(G)$ for a diametral pair	
	$d(u, u, v) = 2d(u, v) \leq \text{tr}(G)$, with equality if and only if	§2
<code>Sec2_readings_Q3</code>	$\text{tr}(G) = 2 \text{diam}(G)$ and $\{u, v\}$ is diametral	
	(Q3) and (Q3') do not change if the triple must consist of	§2
<code>Sec2_readings_Q4</code>	distinct vertices	
	with at least three vertices, the weak and strong readings	§2
<code>Sec2_Q4_imp_Q4'</code> , <code>Sec2_Q3_imp_Q3'</code>	of (Q4) agree, and so do those of (Q4')	
<code>Sec2_classReading</code>	(Q4) implies (Q4'); (Q3) implies (Q3')	§2
	neither “every distance-hereditary graph has (Q3’)” nor	§2
	“every one has (Q4)” holds	
<code>Sec2_problem2_altReading</code>	not every median graph has (Q3)	§2
<code>Lemma1</code>	Lemma 1	§3
<code>Sec3_monotonePath</code>	monotone lattice paths give condition (a)	§3
<code>TheoremA_grid</code> , <code>TheoremB_grid</code>	$G_1 \cong \Gamma(S_1)$ and $G_2 \cong \Gamma(S_2)$ for the placements of	§3
	Section 3, and S_1 , S_2 satisfy (a) and (b)	
<code>TheoremA_i</code> , <code>TheoremA_ii</code>	Theorem A except its last sentence, with both definitions	Thm A
	of “peripheral”, and the further facts of its proof; G_1 lacks	
	(Q3) and has (Q4) in both readings	
<code>TheoremB_i</code> , <code>TheoremB_ii</code>	Theorem B except its last sentence (“only triametral	Thm B
	triple”: every triametral triple consists of 1, 4, 7), and the	
	facts of its proof; G_2 lacks (Q4) and (Q4') in both	
	readings and has (Q3)	

Lean statement	what it says	here
Sec3_medianDH_noQ4	some graph is median and distance-hereditary and lacks (Q4)	§3
Sec3_DH_noQ4_imp_Q3	a distance-hereditary graph that lacks (Q4) has (Q3)	§3
Sec3_M0_notDH	the example of [13] is not distance-hereditary, with the induced path given there	§3
TheoremA_min, TheoremA_unique	the last sentence of Theorem A	Thm A
TheoremB_min, TheoremB_unique	the last sentence of Theorem B, for (Q4) and for (Q4')	Thm B
Sec4_median_triangleFree	a median graph has no triangle	§4
Sec4_nonCutVertex	a connected graph with at least two vertices has a vertex whose removal leaves it connected	§4
Sec4_reduction	a median graph with eight vertices has a vertex v such that $G - v$ is connected and triangle-free and the neighbours of v form a nonempty independent set	§4
Sec4_count_triangleFree7	59 classes of connected triangle-free graphs with seven vertices	§4
Sec4_count_candidates	over these 59 classes, 1,857 nonempty independent sets	§4
Sec4_count_median	1, 1, 1, 3, 4, 11, 23, 69 classes of median graphs with 1, ..., 8 vertices	§4
Sec4_count_labelled	1,008,904 labelled median graphs on $\{0, \dots, 7\}$, of which 20,160 lack (Q3') and 20,160 lack (Q4)	§4
Sec4_aut	G_1 and G_2 have two automorphisms each	§4
Sec5_FP_symm, Sec5_FP_iff_viii	FP is symmetric (for natural numbers and for integers); three sums satisfy FP exactly when they satisfy (viii)	§5
BandeltMulder_DH_fourPoint	every finite distance-hereditary graph satisfies the four-point condition [1, Theorem 2]	§5
BandeltMulder_extension	the statement from [1, Theorem 1] in the proof of Theorem B	§3
Lemma2, Proposition3	Lemma 2 (for integers) and Proposition 3	§5
TheoremC_i, TheoremC_ii	Theorem C: the equation, and (Q3) or (Q4) in both readings, (Q3') or (Q4), and (Q3') or (Q4')	Thm C
TheoremC_FP	the equation and the conclusions of Theorem C for finite connected graphs that satisfy the four-point condition	§5
Remark4_i, Remark4_ii, Remark4_iii	Remark 4 (1), (2), (3)	§5
Remark5_i, Remark5_ii, Remark5_iii	(Q4) when $\text{tr}(G) = 2 \text{diam}(G)$; the facts about H_{11} ; not every distance-hereditary graph with $\text{tr}(G) > 2 \text{diam}(G)$ has (Q3')	§5
Sec6_DHtests	G_2 , H_{11} , G and H are distance-hereditary; G_1 and the graph of [6, Figure 2] are not	§5
Sec8_open	G_1 has (Q4) and G_2 has (Q3')	§7
Released_fourPoint_facts	G , H and G_2 satisfy the four-point condition, G_1 does not (a part not used in the text), and G is not median	§5, note 1

How the proofs work. Distances in the concrete graphs are given by tables, identified with Mathlib's distance by a lemma that checks the table against the edges. The distance-heredity of G_2 , H_{11} , G and H is checked from the definition, over all sets of vertices. The implication of Bandelt and Mulder is proved as described in Section 5. The enumeration of Section 4 is carried out in Lean: the proof constructs representatives of the connected triangle-free graphs with at most seven vertices, proves that every such graph, on any vertex type, is isomorphic to one of them, by an explicit isomorphism for each representative with one vertex fewer and each nonempty independent set, and separates the 59 representatives with seven vertices by isomorphism invariants. For the minimality statements, every median graph with at most seven vertices is connected and triangle-free, hence isomorphic to one of these representatives, and for each representative a distance table shows that it is not median or that it has (Q3') and (Q4) (and (Q4) implies (Q4')). For eight vertices, the proof removes a vertex whose removal leaves the graph connected; what remains is isomorphic to one of the 59 representatives, and the removed vertex is adjacent to a nonempty independent set of it, so the graph is isomorphic to one of the

1,857 graphs of Section 4, formed from a representative and one of its nonempty independent sets. For each of these 1,857 graphs the proof gives either three rows of its distance table, which exhibit three vertices with two medians or with none, or an explicit isomorphism onto one of 69 representatives. The 69 representatives are median (by their distance tables), pairwise non-isomorphic (by an isomorphism invariant), and each has (Q3') and (Q4) or is isomorphic to G_1 or to G_2 ; since G_1 has (Q4) and G_2 has (Q3'), this gives the uniqueness statements, and it gives the number 69. The numbers 1, 1, 1, 3, 4, 11, 23 come from classifying the representatives with at most seven vertices in the same way. The number 1,008,904 is the sum of $8!/|\text{Aut}(R)|$ over the 69 representatives R . Each automorphism group is counted by fixing vertices one at a time. Automorphisms given explicitly, which fix the vertices already fixed, move the next vertex to every vertex of a set O ; a colouring of the vertices by their distances to the fixed vertices, refined twice, is preserved by every automorphism that fixes them, and shows that none moves the next vertex outside O . Once all colours are distinct, only the identity is left, so the order of the group is the product of the sizes of the sets O . The two counts $20,160 = 8!/2$ follow from the uniqueness statements, since G_1 and G_2 have two automorphisms each. Finite checks use `decide` and kernel reduction; there is no `native_decide`.

Checks. For every statement of Table 1, the recorded axiom audits (`#print axioms`) report only the axioms `propext`, `Classical.choice` and `Quot.sound`. A static scan found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern`, `unsafe`, and no axiom declaration, in the proof files. The Lean package of this paper, 201 modules (the 199 files of `lean/Research/Backfill/Paper4/` and the two modules of version 1 that they import), was built module by module from an empty directory, reusing only the pinned toolchain and the compiled Mathlib and its dependencies, and every module was checked with `leanchecker`, all with exit code 0, twice (on 28 September 2026 between 18:34 and 20:11 UTC). For the release, the whole Lean project of the repository was rebuilt from a clean directory and all its modules were checked again with `leanchecker` (28 September 2026, 20:48–22:16 UTC). The Lean files are in version 1.9.0 of the repository [3], directory `lean/Research/Backfill/Paper4/`.

Definitions the reader must trust. Apart from Mathlib’s graphs, distances, diameters, eccentricities, connected graphs, induced subgraphs, walks, isomorphisms, cardinalities, cliques (`CliqueFree`) and independent sets (`IsIndepSet`), the statements use the following definitions, each at most eight lines long: `triDist` and `triameter` ($d(u, v, w)$, and its maximum over ordered triples, with repetitions allowed); `IsTriametral`, `IsDiametral`, `IsPeripheral` (eccentricity equal to the diameter) and `IsPeripheralPair` (lying in a diametral pair); `InInterval` and `IsMedian` (connected, and exactly one vertex in the three intervals of any three vertices); the properties `Question3`, `Question3'`, `Question3'Pair`, `Question4`, `Question4'`, their strong and distinct-vertex readings, and the problem claims `Problem1Claim`, `Problem1ClaimPair`, `Problem2Claim` and `Problem2ClaimWeak`; the equation of Theorem C (`TheoremCEquation`); `IsDistanceHereditary`, `IsPendant` and `AreTwins`; `FP`, `FourPointBM`, and the conditions (vii) and (viii) of [1]; the ℓ^1 distance, the graph $\Gamma(S)$, coordinatewise medians, conditions (a) and (b) and monotone walks; the edge lists of G_1 , G_2 , H_{11} , the graphs G and H of [6, Figure 3], the graph of [6, Figure 2] and the graph of [13], the placements of Section 3, and the counting predicate `ClassCount` (a list of pairwise non-isomorphic representatives such that every graph with the property is isomorphic to one of them). A file of known-answer tests checks these definitions, except `ClassCount` and `TheoremCEquation`, on small graphs, including examples where they fail.

Faithfulness and non-claims. A graph is a Mathlib `SimpleGraph` on a finite type, and “a graph with n vertices” means that the type has n elements; the minimality and uniqueness statements therefore cover graphs on every finite vertex type. Following Mathlib, a graph

without vertices is not connected, which is why Lemma 1 needs S to be nonempty. The triameter is a maximum over ordered triples with repetitions allowed, as in [6]. A vertex is peripheral in Lean if its eccentricity equals the diameter; the wording of [6], “lies in a diametral pair”, is a second definition, proved equivalent for connected graphs, and Problem 1 is refuted with either. (Q4) and (Q4′) are stated in their weak reading, which is their definition; the strong readings are separate definitions, and Table 1 says where they are proved or refuted. “Labelled graphs on $\{0, \dots, 7\}$ ” are the graphs on Fin 8. The edge lists of the graphs of [6] and [13] were compared with the sources by the reviewing agents, not by a machine, and the same holds for the agreement of the Lean definitions with those of the paper and of [6].

7 An open question

As G_1 has (Q4) and G_2 has (Q3′), our examples leave open the analogue of Problem 3 for median graphs: does every median graph have (Q3′) or (Q4)? The same question can be asked for modular graphs, in which any three vertices have at least one median; each of (Q3′) and (Q4) fails for some modular graph [6, Section 4].

Tool and computational resource disclosure

Version 1 of this paper was produced on 26 September 2026 (UTC) in a workflow directed by the author, using AI coding agents: Anthropic Claude Code, with the model Claude Opus 5.5. The author set the research programme, namely resolving open problems with AI agents and formal verification, and directed the agents. Claude Code agents selected the question, found the results and their proofs, wrote the verification programs and the partial Lean formalisation of version 1, carried out the literature searches described in the introduction, and drafted the text. An independent Claude Code agent reviewed the results before the paper was written, and a second one refereed the paper: it rechecked every proof, compared the Lean statements with the theorems, checked the MathOverflow credit and the cited theorem of Bandelt and Mulder, and recomputed all counts with its own programs.

Version 2 was produced from 27 to 29 September 2026. A Claude Code agent (model Claude Opus 5.5) wrote the statement file of Section 6, and an independent Claude agent reviewed version 1 of it and recomputed the enumeration with its own programs. OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`) then checked version 1 of the paper independently, recomputing the enumeration with its own program, reviewed version 2 of the statement file and froze it, and wrote the Lean proofs of 46 of the 50 statements, of which it checked 44, batch by batch (compilation from an empty directory, `leanchecker` and `axiom audit`). When Codex’s usage quota ran out, Claude Code compiled and checked Codex’s proofs of the two minimality statements, and wrote and checked the proofs of the other four statements (the uniqueness statements of Theorems A and B, and the counts 1, 1, 1, 3, 4, 11, 23, 69 and 1,008,904, with 20,160, of Section 4), building on Codex’s enumeration; independent Claude agents reviewed these six proofs. Claude Code also assembled the Lean package of Section 6 and ran the checks described there. Writing and reviewing the statement file brought to light the three missing hypotheses that version 2 adds. Claude Code drafted the changes of version 2, and another independent Claude Code agent refereed a draft of version 2. Codex has not reviewed version 2 of the paper or the Lean proofs written by Claude Code; the final review before release was done by an independent Claude Code agent in its place. Throughout, “we” refers to this workflow. The author is not a professional mathematician. The author has read version 2 of the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 1 and on the

agreement of those statements with the paper, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

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